Homework 7

Due: Friday, April 8

All homeworks are due at 11:59 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

LaTeX tips

Here are some LaTeX commands that might help you

`\binom{a}{b}` or `{a \choose b}`	$\binom{a}{b}$
`\sum_{i=0}^{n} a_i`	$\sum_{i=0}^{n} a_i$

LaTeX *is* required. You should be using the `cs22` document class. Look for a LaTeX guide here for more information.

Problem 1

Assume there are 101 hot drinks, each of which has some nonnegative, integer number of marshmallows. Prove that it is possible to choose 11 of them whose total number of marshmallows is divisible by 11.

Hint: Associate each hot drink to its number of marshmallows modulo 11, and consider two cases: one where there is a hot drink in each of the categories and one where there isn't.

Problem 2

Consider the following equation:

$$x_1 + x_2 + x_3 + x_4 = 75$$

where each x_i must be a non-negative integer.

a. Count the number of solutions to this equation.

b. Now suppose we require a solution with x_1 and x_3 strictly positive. Count the number of solutions under this new constraint.

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c. More generally, suppose we require a solution where some a_i are fixed constant nonnegative integers and $x_i \geq a_i$ (that is, we have a constant a_i corresponding to each x_i), satisfying

$$\sum_{i=1}^{4} x_i \leq 75.$$

That is, each x_i is restricted to be *at least* a_i. Again, count the number of solutions under this new constraint. You can give this as an expression in terms of the a_i's. You may use summation (Σ) notation in your final answer to express this count.

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Problem 3

Prove the following equality.

Do not use any algebraic manipulation in your argument. Instead, give a "counting argument": why is the number of ways to chose k objects from n options the same as the sum shown on the right?

$$\binom{n}{k} = \binom{n-2}{k} + 2\binom{n-2}{k-1} + \binom{n-2}{k-2}$$

☕ Problem 4 (Mind Bender — *Extra Credit*)

Mind Benders are extra credit problems intended to be more challenging than usual homework problems and are an exploration into a topic not covered in lecture. This week, we have an exploration into *generating functions*.

We define a sequence of numbers as a list of (indexed) numbers:

$$a_0, a_1, a_2, a_3, a_4, \ldots$$

A sequence can be finite[1] or infinite, can have repeats, and need not be in increasing or decreasing order.

Example

Here are some sequences:

i. The constant 1 sequence: $1, 1, 1, 1, 1, \ldots$

ii. The sequence of positive integers: $1, 2, 3, 4, 5, \ldots$

iii. The Fibonnaci numbers: $1, 1, 2, 3, 5, 8, \ldots$

iv. $\binom{6}{i}$ where $a_i = \binom{6}{i}$:

a_0,	a_1,	a_2,	a_3,	a_4,	a_5,	a_6.
$\binom{6}{0}$,	$\binom{6}{1}$,	$\binom{6}{2}$,	$\binom{6}{3}$,	$\binom{6}{4}$,	$\binom{6}{5}$,	$\binom{6}{6}$.
1,	6,	15,	20,	15,	6,	1.

Generating functions are functions (in x) that generate a sequence as the coefficients of the powers of x.

Example

For example,

$$\begin{aligned} f(x) &= 3 + 2x + 3x^2 + x^4 \\ &= 3x^0 + 2x^1 + 3x^2 + 0x^3 + 1x^4 + 0x^5 + 0x^6 + \cdots \end{aligned}$$

is the generating function for the sequence $3, 2, 3, 0, 1\ (, 0, 0, \ldots)$.

[1] If a sequence is finite, we assume there are infinitely many zeros at the end.

In general,

$$f(x) = \sum_{i=0}^{n} a_i x^i$$

is the generating function for sequence $a_0, a_1, a_2, \ldots$.[2]

a. Using the binomial theorem, verify that the function

$$b(x) = (1+x)^6$$

generates sequence (iv) from the example above. That is, it generates the sequence $a_0, a_1, a_2, \ldots$ where $a_i = \binom{6}{i}$.

Can you generalize this if you replace 6 with an arbitrary n?

b. Let's say our sequence a_i is the *number of ways* to make i cents with 2 one-cent coins, 3 nickels, 2 dimes, and 1 quarter. It starts as[3]

a_0,	a_1,	a_2,	a_3,	a_4,	a_5,	$a_6 \ldots$
1,	1,	1,	0,	0,	1,	$1 \ldots$

Justify that

$$f(x) = (1+x+x^2)(1+x^5+x^{10}+x^{15})(1+x^{10}+x^{20})(1+x^{25})$$

is the generating function for this sequence.

How many ways can we make 26 cents? Use a tool like *WolframAlpha* to expand the polynomial and check that the coefficient of x^{26} in the polynomial is what you got.

c. The generating function for sequence (i) from above is[4]

$$f(x) = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$$

Consider the generating function

$$g(x) = \frac{1}{(1-x)^4} = (1 + x + x^2 + x^3 + \cdots)^4$$

Use a computer to (Taylor-series) expand this function as a polynomial. Explain why the coefficient of x^{75} gives the solution to 2a from this homework.

Then, count the number of solutions to $x_1 + x_2 + x_3 = 50$ (where each x_i is a nonnegative integer) using generating functions.

[2] Check that this makes sense before beginning.

[3] There is only one way to make 0 cents (no coins at all). There is 1 way each to make 1 and 2 cents (we use the one-cent coins); we can't make 3 or 4 cents with our coins; we can make 5 cents with a nickel; we can make 6 cents with nickel plus one-cent.

[4] This comes from the Taylor series expansion of f about 0.